8 Sep 2026 (Tuesday).

EXCRETORY SYSTEM

Excretion → Removal of nitrogen containing (nitrogenous) waste products from the body that are produced during metabolic activities in animals.

Nitrogenous waste products.
(i) Ammonia (NH_3)
(ii) Urea ($NH_2-CO-NH_2$)
(iii) Uric acid ($C_5N_4H_4O_3$)
(iv) Creatinine ($C_4H_7N_3O$)

→ Based on excretory substances, organisms are divided into three categories

① Amonotelic → main excretory substance → Ammonia.

Eg → (a) Bony fishes
(b) Amoeba
(c) Paramaecium
(d) Sponge
(e) Crocodile
(f) Tadpole
(g) Protozoa
(h) Salamander

② Ureotalic → main excretory substance → Urea.

Eg. (a) mammals
(b) Amphibians
(c) Aquatic reptiles → Tortoise, Gharial
(d) Cartilaginous fishes) (e) Shark

Note → Earthworm and Ascaris come under both ureotelic and ammonotelic categories.

③ Uricotelic → main excretory product
↳ Uric acid.

Eg → (a) Birds
(b) Insects → cockroach
(c) Reptiles
↳ (Snake, lizard)

Note → Spiders and scorpions excrete guamine, so they are called guanotelic.

Excretory organs in human
① Liver → Production of urea
② Skin → Excretion of sweat
③ Lungs → Excreation of CO_2
④ Kidney → Excretion of urea.

→ Our main excretory substance → Urea / Urine.

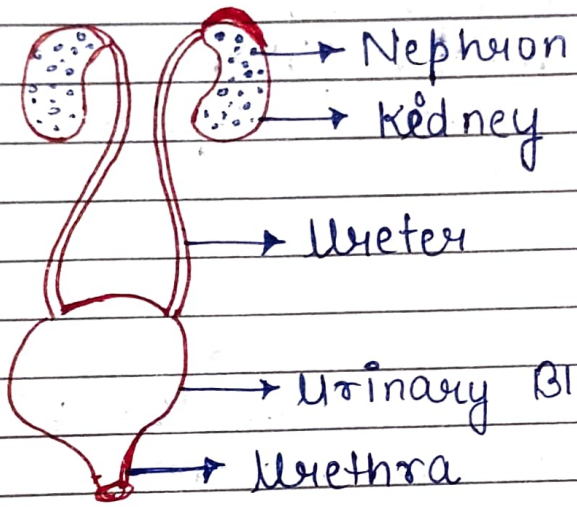
Composition of urine → (i) water – 95%
(ii) Urea – 2.6%
(iii) Uric acid 0.4%
(iv) Cretinine – 2%

- PH of urine → 6 – 6.5
- Nature → Acidic

- colour → Yellow (due to presence of urochrome pigment).
- odour → due to the presence of ammonia.
- Quantity → 1.5-2 litre / day.

❀ 9 Sep. 2026 (Wednesday)

Excretory System



→ There are 4 major structures in the human excretory system.

(i) Kidney (2)
(ii) Ureter (2)
(iii) Urinary bladder (1)
(iv) Urethra (1)

(i) Kidney

→ Our main excretory organ
→ Located in the posterior part of the abdominal cavity.
→ Colour → Deep - Redish brown
→ Number → 2

→ Shape → Like a bean seed
→ weight → males → 150gm
→ females → 135 gm

→ Func^{n} → filteration of blood
→ The kidney func^{n} is based on the process of Ultrafiltration.
→ Cell of kidney → nephron
→ Study of nephron → Nephrology
→ No. of nephron → 1 million / kidney
→ Unit of excretory system → Nephron

(ii) Ureter
→ Number → 2
→ length → 20-25 cm
→ func^{n} → To carry urine from the kidneys to the urinary bladder.

(iii) Urinary bladder
→ A muscular, sac like structure. in which urine is gradually stored.
→ "Detrusor muscles" are found in it.

→ Capacity → In females → 500 ml
→ In males → 700 ml

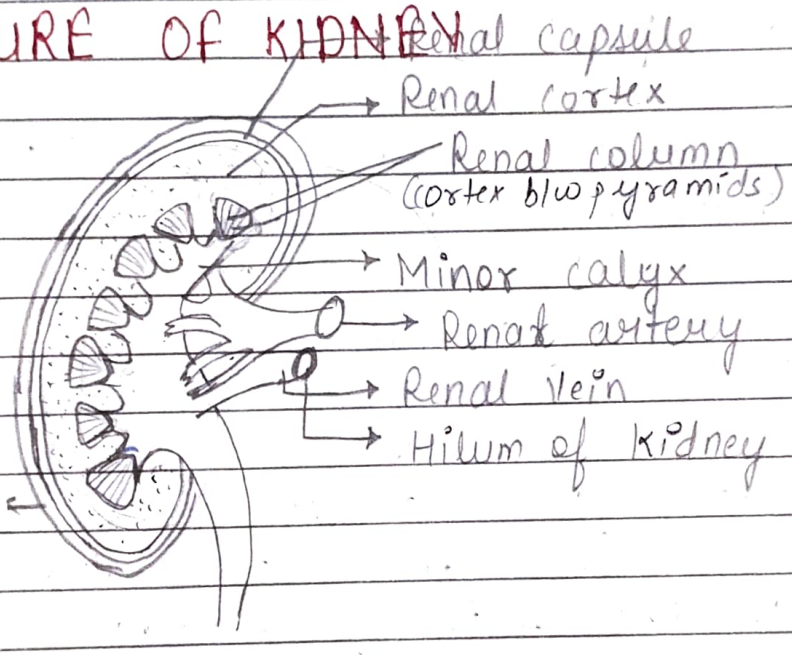
→ The urge (feelings) to urinat occurs once 150-200 ml of urine is accumulated.

(iv) Urethra

→ A tubular part that carries urine from the urinary bladder to the outside of the body.

→ Length → In females → 4-5 cm
In males → 15-20 cm

STRUCTURE OF KIDNEY



→ A renal capsule is found around the kidneys, which protects it from infection.
→ Outside the renal capsule, an adipose capsule is found which is made of fat.
→ It protects the kidney from external injuries and shocks.
→ Outside the adipose layer, the renal fascia is found, which connects the kidney to other structures in the abdominal cavity.
→ The internal part of the kidney is divided into two regions.
(i) Outer dark colour region → Medulla

(ii) Inner light coloured region → Medulla.

→ Medullary pyramids are found in the medulla.

→ These are connected to the calyx.

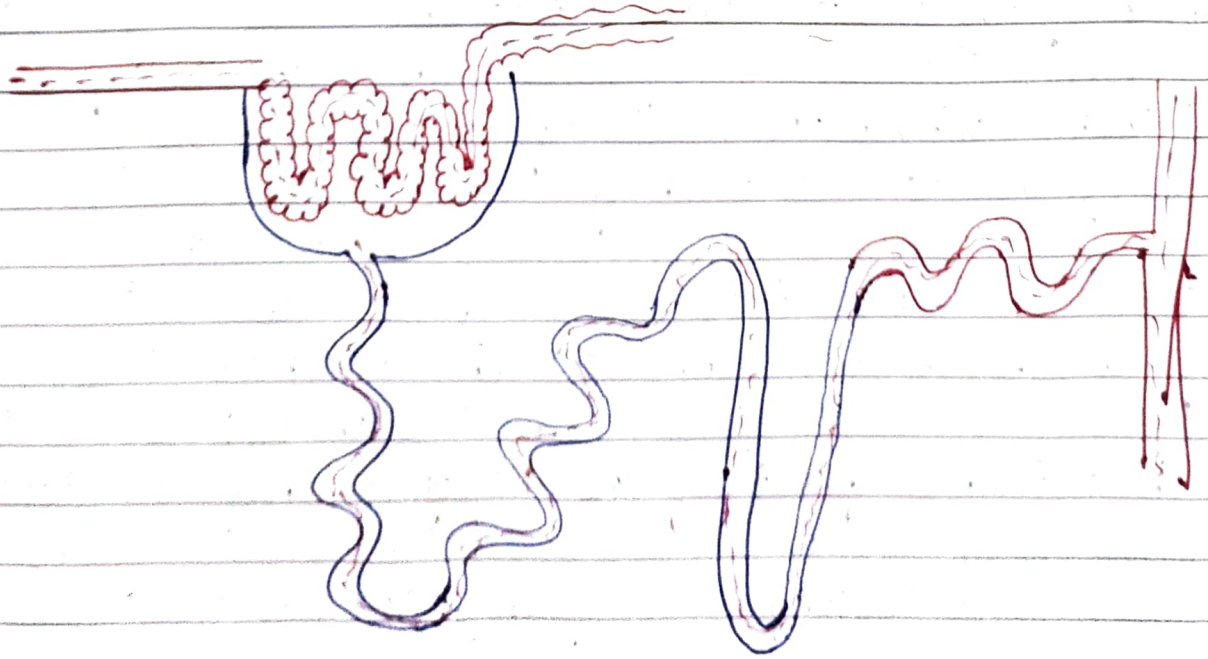
→ The calyx connects to the hilum, which is the central part of the kidney, in the form of branches

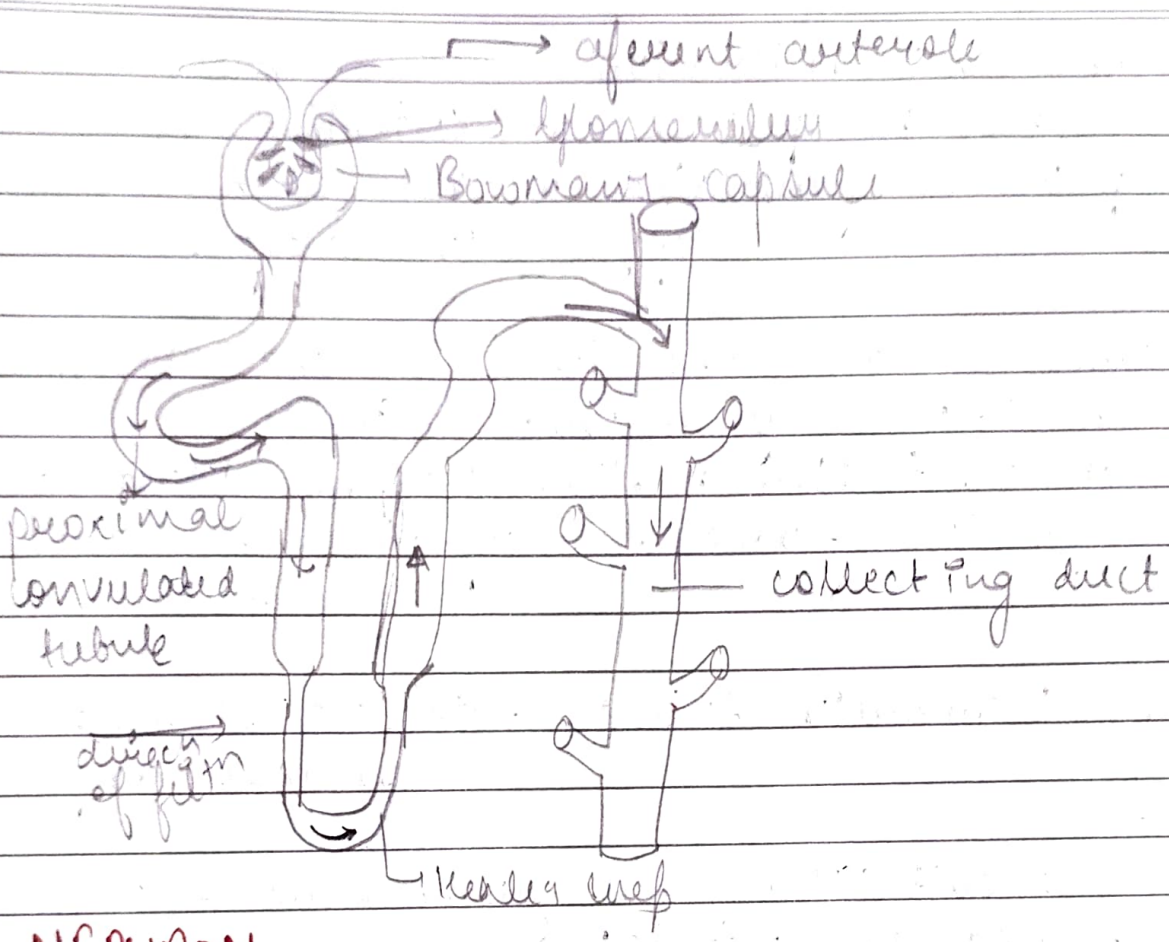
→ The renal artery enters through the hilum, region bringing clean (oxygenated blood).

→ The renal vein also emerges from the hilum, carry impure (deoxygenated) blood away from the kidney.

→ The ureter also exists from the hilum, carrying urine ~~and~~ out of the kidney and into the urinary bladder.

STRUCTURE OF NEPHRON





NEPHRON

↳ Structural and functional unit of kidney
↳ It consists of two main structures.
  ↳ (i) Malpighian corpuscle
  (ii) Renal Tubules.

→ The glomerulus and bowman's capsule are collectively called the malpighian corpuscle

→ There are three process that occur in nephron
(i) Glomerulus filtration
(ii) Tubular reabsorption
(iii) Tubular secretion

(1) Glomerulus filteration

↳ In this the process of filtration due to the pressure generated in the glomerulus (approx. 10 mm Hg).

→ Because of this, the fluid reaches the PCT in the form of glomerular filtrate.

→ In an adult human, 1100 - 1200 ml blood flows through the glomerulus of these nephrons per minute.

→ GFR [Glomerulus filtrate rate] → 125 ml/min or 180 L/day

→ As a result 125 ml of glomerulus filtrate is formed per minute.
↳ per day → 180 liters.

→ This glomerulus filtrate contains water, glucose, amino acids, vitamins, and mineral salts.

(2) Tubular Reabsorption

↳ In the PCT, 65% of this filtrate is reabsorbed.

→ Here water, glucose, vitamins, amino acids and ions are absorbed.

→ In the descending limb of loop of Henle, water and ions are absorbed.

→ The ascending limb of the loop of henle is impermeable to water, Therefore water absorption does not occur in it.

12 Sep 2026 (Saturday)

③ Tubular Secretion

→ Waste substance from the vasa recta vessels surrounding the nephron are directly secreted into the tubules of the nephron.

→ At the end of the above three steps, the fluid present in the collecting duct is now called urine.

Animals	Excretory organs
(i) Unicellular organisms (Amoeba, paramecium)	Cell membrane
(ii) Sponges, Hydra	General body surface
(iii) flatworms ↳ roundworms	Flame cells
(iv) Insects [cockroach]	Malpighian corpuscle
(v) Earthworm	Chloragogan cell
(vi) Mollusca (oyster, snail, octopus)	Organ of Bojanus
(vii) Chordata	Kidney
(viii) Ascaris (Roundworms)	H-shaped Renette cell

- Disease related to excretion

→ pus in urine → pyuria (यूरीन के साथ मवाद आना)

→ Presence of glucose in urine → Glycosuria.

→ Presence of RBC / blood in urine → Hematouria.

→ Increase in the volume of urine → polyuria

→ Burning sensation during urination → Dysuria

→ Increase in the amount of urea in the blood → Uremia.

→ Normal quantity of urea in blood → 20-30mg / 100 ml.

→ When the kidney does not func^{n} properly, the amount of urea in the blood increases. This is called Uremia.

→ In such a situation, purification of blood is done by artificial methods, which is called artificial dialysis.

Discovered by (Machine) → Dr. Murray

Process discovered by → Willeam Johan Kolff.